SDC HACKATHON 2026 DOMAIN 1

SHIFTATLAS 3D

A source-traceable career atlas for Malaysia



Primary: SDG 4
Supporting: SDGs 8 and 10

SELADEVS UTHM

Universiti Tun Hussein Onn Malaysia

The next learning decision

Which skill should Aina learn next?

A fictional Malaysian graduate

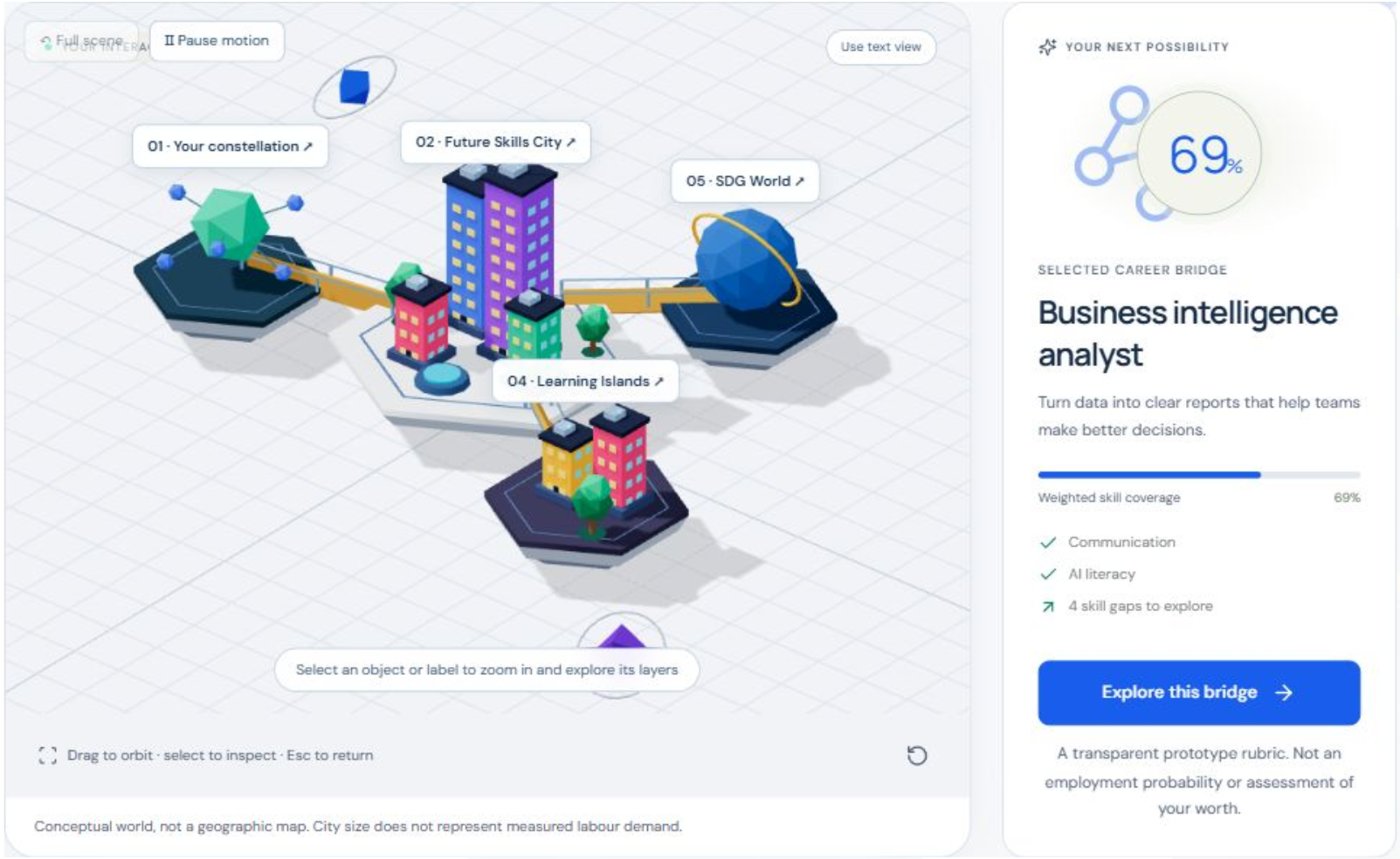
SQL or Python next?
Limited weekly time. Limited budget.

An ordered plan with reasoning she can inspect.



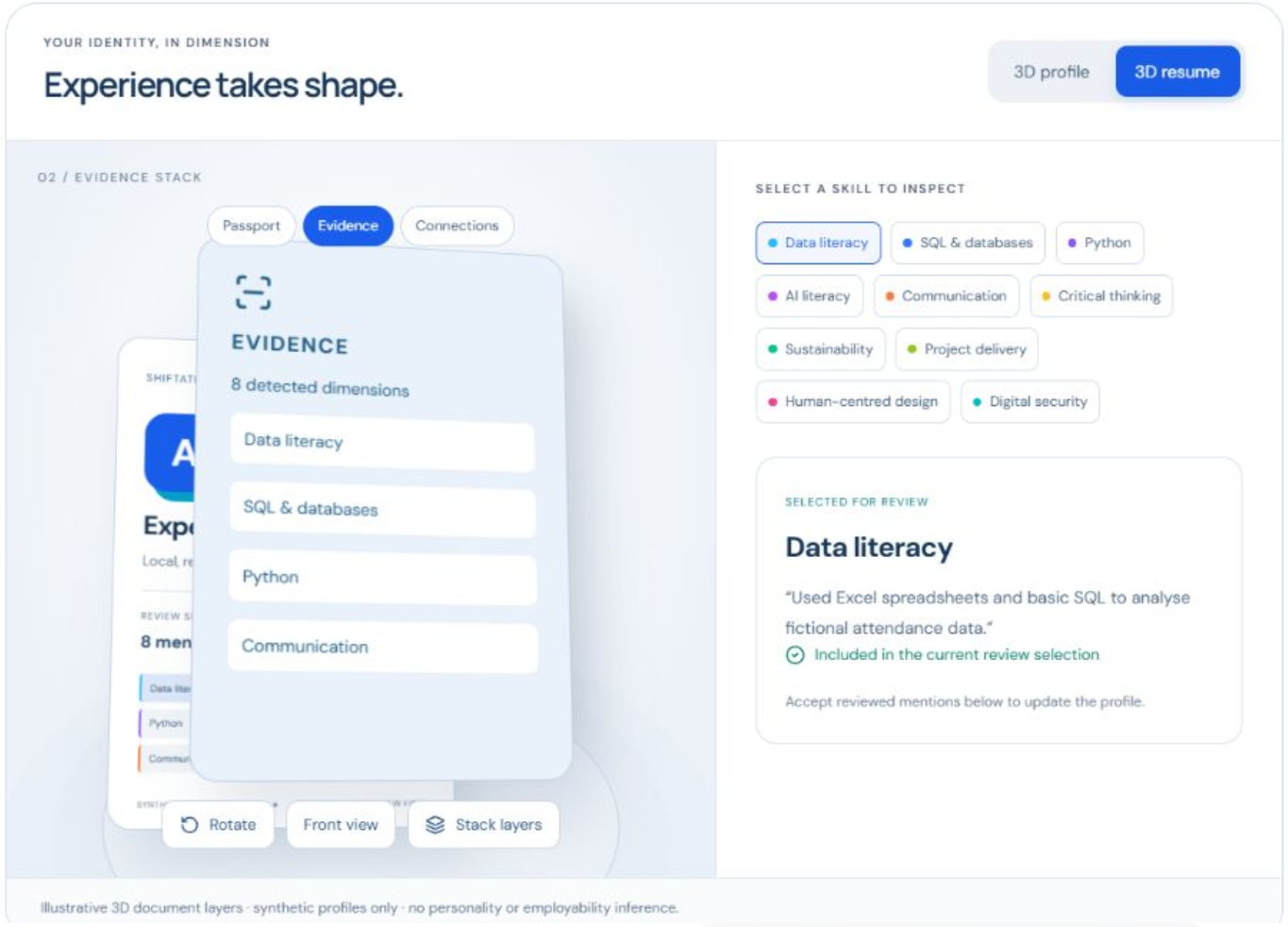
Concept illustration · fictional learner

One profile across six connected modules



Reviewed skills **Dated signals** **Role coverage** **Learning** **SDGs** **Scenarios**

Reviewed intake and a 3D profile



Local parsing

Synthetic PDF, DOCX or TXT

Original evidence

Excerpts remain visible for human review

The user chooses skill levels.
A mention cannot prove proficiency.

Dated signals feed the rubric

69% → 70%

Analyst coverage
in this synthetic example

Six dated research titles
Explicit opt-in weighting
25% maximum weight uplift



Research attention is experimental. Validated demand forecasting remains future work.

Explainable coverage and visible gaps

6 reference roles



10 skill dimensions

Skill	Your level	Target	Base → effective weight	Score contribution	Gap
Data literacy	3/5	4/5	3 → 3.000	18.8 points	1 level
SQL & databases	1/5	3/5	3 → 3.000	8.3 points	2 levels
Python	1/5	2/5	1 → 1.000	4.2 points	1 level
Communication	4/5	3/5	2 → 2.000	16.7 points	Covered
Critical thinking	3/5	4/5	2 → 2.000	12.5 points	1 level
AI literacy	2/5	2/5	1 → 1.000	8.3 points	Covered

Current level · target · weight
Every score contribution stays visible.

O*NET references; project-authored rubric. Malaysian calibration remains open.

Learning in prerequisite order

An affordable next step

Foundations first
Estimated practice hours
Real provider resources
Exportable plan



Activity completion does not certify proficiency. Practice hours are authored estimates.

Malaysia and the 17 SDGs

16 states and federal territories



DOSM observations: 2026 Q2

17 target-level pathways



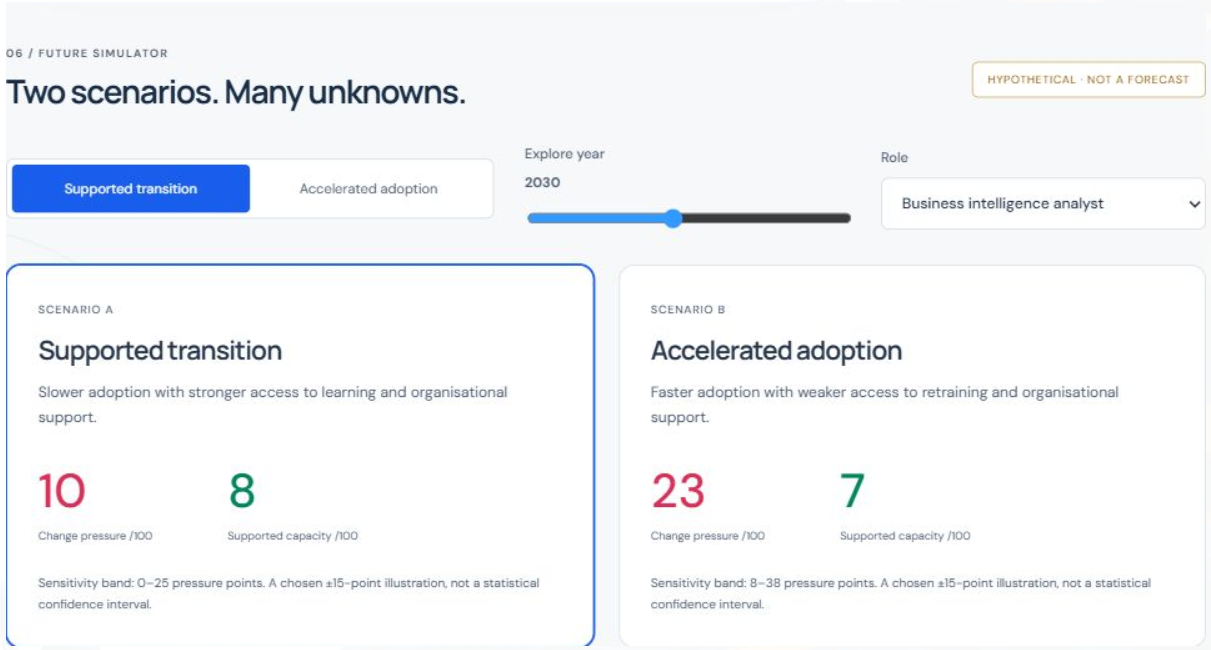


Targets 4.4 · 8.5 · 10.2

Intended contributions; outcomes unmeasured

Two uncertain adoption scenarios

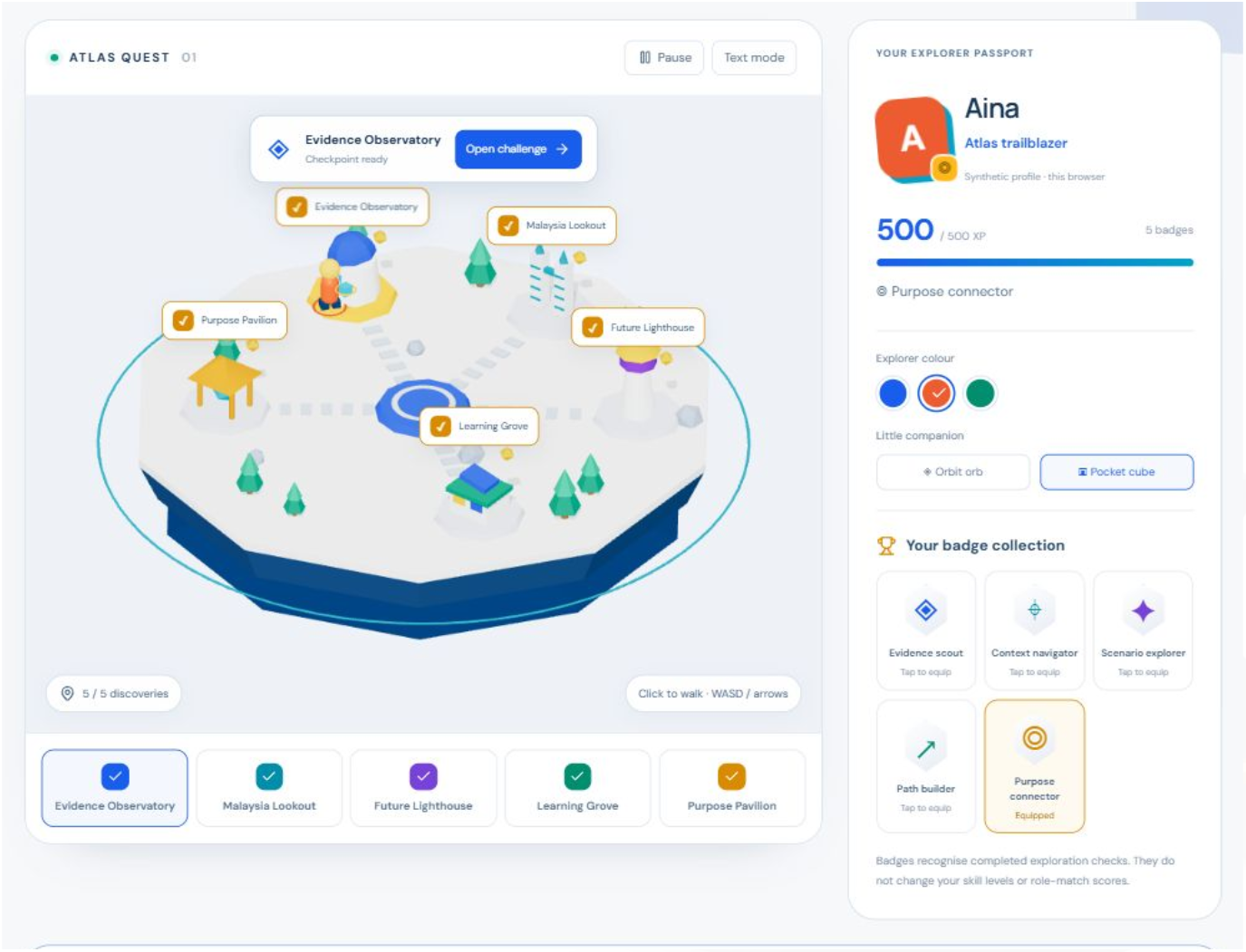




Supported transition and accelerated adoption

2026–2035 · hypothetical indices · disclosed assumptions · no assigned probabilities

Atlas Quest makes exploration playable

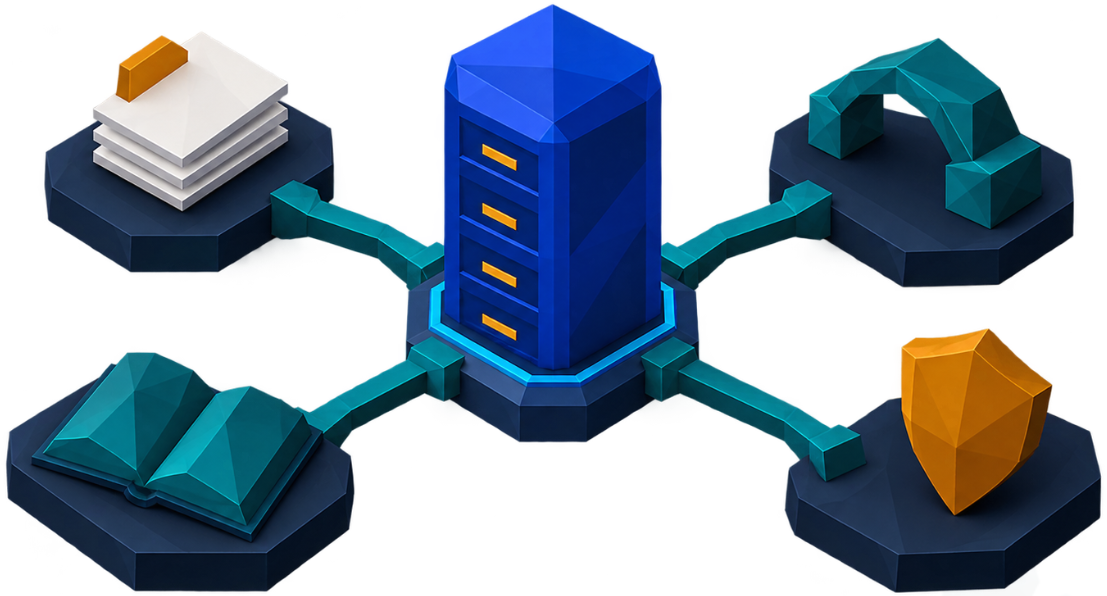


5 discoveries

Unique badges
Saved passport
Outfits and companions

Activity badges.
Scores unchanged.

Career Council: four bounded reviews



Architecture concept illustration

One model. Four perspectives.

01 **Profile reviewer**
02 **Role explorer**
03 **Learning planner**
04 **Grounding critic**

Replit coordinates four sequential RunPod jobs.
The original skill rubric stays unchanged.

Reference-linked reviews · original skill rubric retained

Grounding checks before the answer



01 **Fictional text**

02 **Replit gateway**

03 **RunPod inference**

04 **Validated report or visible failure**

Schema, known IDs and exact quotes checked.
Secrets stay on the server.

Qwen3-4B-Instruct-2507 · Apache 2.0

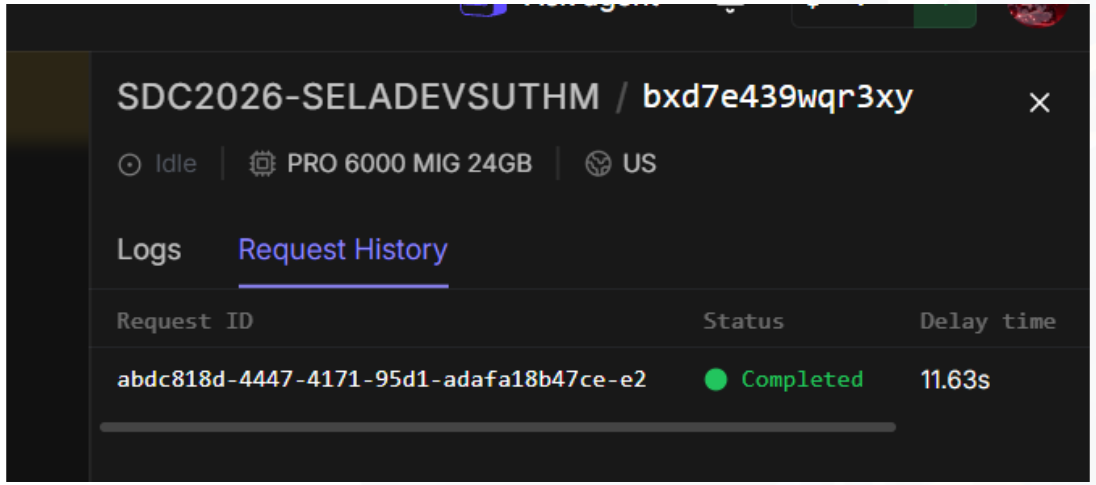
Explicit cloud consent. A critic and valid citations do not verify every generated claim.

Verified E5 GPU execution

3.452 s

Model execution time

PRO 6000 MIG 24GB
CUDA · Runpod Serverless



1 fictional query · 2 authored passages
Queue delay: 11.626 s

multilingual-e5-base

25 October 2026
03:11 MYT



55 application tests · 18 offline worker checks
Production build passed

Single synthetic smoke test. Retrieval quality and Council execution require separate validation.

Next: a learner pilot

Proposed pilot with 10 learners

20% Less median planning time versus a static resource list

80% Correctly explain a score and its uncertainty



Targets only. No learner outcomes measured yet.

Proposed evaluation · learning access, planning clarity and informed career choices